PRAVEENA N

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Machine Learning enthusiast with hands-on experience building LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, Generative AI workflows, and backend services. Experienced in developing AI solutions using Python, FastAPI, SQL, and modern machine learning frameworks. Passionate about transforming research ideas into production-ready AI systems and solving complex real-world challenges through experimentation and innovation.

PROJECTS

Job discovery platform **June 2026**

- Built a resume parsing service that ingests multiple file formats (PDF, DOCX, and more) and extracts structured candidate data using an LLM-based extraction pipeline.
- Exposed a full CRUD API so candidates can review and edit their extracted profile before it's used downstream.
- Architected a 7-stage async pipeline covering query generation, search, and ranking to power live job discovery.
- Integrated dual-source job search Adzuna API and a SearXNG/crawl4ai web-scraping fallback to maximize live job coverage.
- Implemented hybrid ranking with LLM-based candidate-fit judging to surface the best-matched job listings.

Tech Stack: Python, FastAPI, LLMs, GenAI, Prompt Engineering, Postergres, REST APIs, GitHub.

RAG-Based Technical Assistant **March 2026**

- Developed a Retrieval-Augmented Generation (RAG) based Technical Assistant capable of answering technical queries using real-time information from community discussions and technical documentation.
- Built automated web scraping pipelines to collect and process content from Discourse forums, creating a continuously updated knowledge repository.
- Implemented document chunking, embedding generation, semantic search, and retrieval pipelines to provide context-aware responses.
- Integrated Large Language Models (LLMs) to generate grounded and accurate answers from retrieved knowledge.
- Optimized retrieval performance through metadata filtering, prompt engineering, and embedding-based similarity search.

Tech Stack: Python, Web Scraping, Embeddings, ChromaDB, Cosine Similarity, LLMs, GenAI

SKILLS

Python, SQL, Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, FastAPI, TensorFlow, PyTorch, Scikit-learn, Vector Databases, Embeddings, Semantic Search, Web Scraping, REST APIs, AWS, MySQL, PostgreSQL, MongoDB, Git, Docker, Pandas, NumPy, Data Analysis, Feature Engineering, Model Evaluation, Predictive Analytics, PowerBI, Tableau.

PROFESSIONAL EXPERIENCE

Boeing India Private limited

Software Development Executive Apprentice **Aug 2025 - Aug 2026**

- Integrated AI with MCP services, GitLab, databases, and VS Code to automate software testing workflows.
- Worked on testing two applications and reported over 50 bugs, improving software quality and reliability.
- Leveraged PyTesseract for OCR-based computer vision to extract data from images and integrated it with Selenium to automate end-to-end testing workflows.

EDUCATION

- Bangalore Institute of Technology | BTech | ISE **8.8 | Nov 2022 - July 2026**
- IIT Madras | BS | Data Science **7.89 | Sept 2022- Dec 2026**

ACHIEVEMENTS

- Best Capstone Project Award – **IIT Madras** for GenAI-based software engineering project.
- Completed IIT Madras ML workshop on implementing machine learning algorithms using NumPy.